



CO3-AT1-Debugging and Optimization

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QUESTIONS

Q1ATM Withdrawal – Exception
Handling Debugging
10 marks**Q2**Hospital Registration –
NullPointerException
Debugging
10 marks**Q3**E-Commerce Product Search –
Array Exception Debugging
10 marks**Q4**Railway Reservation –
Synchronization Debugging
10 marks**Q5**Bank Account – Race Condition
Debugging

Q2 Hospital Registration – NullPointerException Debugging

10 marks

A hospital patient registration application crashes when patient information is unavailable.

Debug the following complete Java program.

Task:

1. Identify the runtime error.
2. Modify the program to handle a missing patient name.
3. Handle invalid age input.
4. Use try-catch-finally appropriately.
5. The program must always display "Registration completed".
6. Submit the COMPLETE corrected Java program.

Faulty Program:

```
import java.util.Scanner;

class PatientRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
System.out.print("Enter patient name: ");
String inputName = sc.nextLine();

String name;

if (inputName.equals("NULL")) {
    name = null;
} else {
    name = inputName;
}

System.out.print("Enter age: ");
int age = sc.nextInt();

System.out.println("Patient Name: "
    + name.toUpperCase());

if (age >= 18) {
    System.out.println("Adult patient");
} else {
    System.out.println("Minor patient");
}

System.out.println("Registration completed");
}
}
```

The corrected program must prevent NullPointerException and handle invalid age input without terminating unexpectedly.

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 class PatientRegistration {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String inputName = sc.nextLine();
6         String name;
7         if(inputName.equals("NULL"))
8             name = "Unknown";
9         else
10            name = inputName;
11        int age = sc.nextInt();
12        System.out.println("Patient Name: "+ name);
13        if(age>=18)
14            System.out.println("Adult Patient");
15        else
16            System.out.println("Minor patient");
17        System.out.println("Registration Completed")
18    }
19 }
```

 Submitted

Test Input

NULL
18

Output

Patient Name: UNKNOWN
Adult Patient
Registration Completed

 Pending manual review